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Work-Based Learning and Early Career Outcomes: Evidence on Earnings, College-Level Employment, and Career Progression

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Abstract
Prior research has linked internships to career benefits for students. But are the benefits we see due to the internship itself or to other characteristics of students that make them both more likely to secure an internship and more likely to get a higher paying job after graduation? And do other forms of work-based learning also convey career benefits? In this study, we use a matching design on restricted-access data from the National Center for Education Statistics to analyze the link between work-based learning and employment outcomes for bachelor's degree completers one and four years after graduation. Work-based learning pathways explored include paid and unpaid internships, co-ops, practica and federal work-study jobs. After pre-processing the data via nearest neighbor matching on student academic and demographic characteristics to approximate experimental conditions, we find evidence that paid internships are associated with higher earnings one and four years after completion, as well as higher likelihood of college-level employment and career progression. We hypothesize three key mechanisms by which work-based learning might improve employment outcomes: development of transferable skills, networking that facilitates access to professional connections, and resume signaling for employers that demonstrates otherwise unobservable qualities of a candidate. Our analysis provides some support to all three mechanisms, but the strongest case for the third mechanism: symbolic capital/ resume signaling. Unpaid internships and practica experiences are linked to higher likelihood of college-level employment and career progression, but not to earnings benefits.

Keywords Work-based learning · Employment outcomes · NCES · College internships

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Introduction

Work-based learning is a concept older than postsecondary education. For centuries, apprenticeship programs provided the means for a learner to get hands-on training under the direction of an experienced master craftsman to learn a trade and enter a profession (Gamble, 2001). A college education, on the other hand, was first seen as an alternative to learning a trade. Courses of study were deliberately theoretical rather than applied, to train the learner in how to wrestle with large concepts and ideas, and to pass on and contribute to a shared intellectual tradition, but not necessarily to gain knowledge that would prepare a student for a particular job upon graduation (Chan, 2016). The theoretical and the practical have sometimes been at odds, with higher education purists questioning the value and place of curricula that are too closely tied to current labor market needs lest this compromise the quality and sustaining value of a college degree.

Students tend not to get caught up in the either/or debate. Survey evidence shows that students value higher education both for its potential to help them become the best version of themselves as well as to succeed in a rewarding and stable career (Armstrong & Hamilton, 2013; Chan, 2016). Given the substantial investment of time and money that students and families make to attend higher education, the latter goal is essential, not optional, especially for students coming from families with lower income and wealth.

There is good evidence that a college education brings substantial social and financial rewards. Yet the likelihood of experiencing these rewards is uneven (Armstrong & Hamilton, 2013). For some students, the investment pays off many times over, but for others it comes up short. One promising practice to mitigate the risk and improve the value of higher education is to combine a traditional college education with the even older traditional form of learning—learning through work (Lester & Costley, 2010). This allows a student to apply and deepen what they are learning in the classroom via a structured and related experience in the workplace (Lester & Costley, 2010). Work-based learning refers to experiences that allow students to connect academic learning with real-world experiences via internships, practica, work-study, and co-ops. It is important to distinguish these structured work-based learning experiences from general employment. Many students work while enrolled in college, yet most paid jobs are not explicitly designed to support career or professional development, provide mentorship, or offer industry aligned task mastery. In contrast, effective work-based learning experiences typically include intentional learning goals and tasks aligned with those goals, supervision from professionals in the field, professional development opportunities to enhance career readiness, and opportunities to build career relevant networks (e.g., Hora et al., 2024; Just Capital, 2025).

The literature around the career impact of internships and other forms of work-based learning is promising, but nascent. Many of the studies are from outside of the United States or are restricted to a single field of study—most commonly business (e.g., Brook & Corbridge, 2016). Furthermore, there is often no way to untangle whether the better outcomes seen by students who participated in internships are due to the internship itself, or to the fact that students who already had characteristics that made them likely to succeed (e.g., academic success, parents and peers with professional connections, motivation, interpersonal skills) were also more likely to get internships (Bolli et al., 2021; Di Pietro, 2022). Therefore, the following research questions guide our study:

1. In the United States, do students who participate in various types of work-based learning (for example, paid and unpaid internships, co-ops, practica, work-study) experience career benefits?
2. Knowing that participation in work-based learning varies by demographic group, academic preparation, and field of study, how much of the relationship between work-based learning and post-completion outcomes can be attributed to the experience itself as opposed to the characteristics of the students who participate?
3. *Why* might certain types of work-based learning be associated with stronger career outcomes?

This study helps to push the boundaries of our knowledge around the career outcomes related to internships and other forms of work-based learning by employing a matching design that helps to neutralize the selection effect of some of these student characteristics. The longitudinal nature of the data allows us to examine the relationship of work-based learning experiences to career outcomes not only immediately after graduation, but also four years later. This study also makes a unique contribution by looking across the spectrum of four types of work-based learning experiences: internships, co-ops, practica, and federal work-study.

The following section outlines the current state of knowledge around the benefits of internships and other forms of work-based learning and the mechanisms by which these benefits might be conveyed. Next, we detail the data and methods used to conduct our analysis, followed by a discussion of the results and suggestions for further research.

Evidence on Work-Based Learning And Career Outcomes

There is consistent evidence that internships are associated with career benefits; what is not well understood is why this relationship exists and whether it extends to other modes of work-based learning apart from paid internships. Existing research has documented that college graduates with internship experience see a range of employment advantages compared to their peers without such experience. These graduates receive more interview invitations, are less likely to be unemployed during the crucial first post-graduation year, experience better alignment between academic fields and employment sectors, earn more money, and report greater career satisfaction (e.g., Di Meglio et al., 2022; Jung & Lee, 2017; Margaryan et al., 2022; Nunley et al., 2016; Silva et al., 2018).

These enhanced employment outcomes may operate through three key mechanisms (e.g., Becker, 1964; Granovetter, 1973; Spence, 1973): development of skills that are transferable across occupations (human capital); networking that facilitates access to professional connections (social capital); and resume signaling that employers use as a proxy for otherwise unobservable candidate qualities (symbolic capital).

The first mechanism through which internships enhance employment outcomes may be akin to the way in which other forms of education or work experience are expected to improve productivity and employment outcomes: the development of skills, also known as human capital (Becker, 1964; Mincer, 1974; Schultz, 1961). Evidence demonstrates that internships help facilitate skill development, including inter-personal and intercultural skills, collaboration, self-confidence, and professional readiness (Benbow et al., 2021; Di Pietro, 2022; Lambert Snodgrass et al., 2021; Sekiguchi et al., 2023). Beyond the theoretical link, there is

evidence from following the career trajectories of interns in Switzerland that the primary income advantage for interns comes from developing broadly applicable skills rather than narrow organizational knowledge (Bolli et al., 2021).

Second, internships may influence employment outcomes by improving social capital. Research has demonstrated that social capital–in the form of access to relationships and the ability to leverage them—is linked to enhanced employment opportunities (Granovetter, 1973; Lin, 1999). Accessing and leveraging social capital outside of one's immediate family and social circles is especially important for individuals facing structural disadvantages in the labor market, including women and racial and ethnic minorities, who often encounter discrimination and unequal access to opportunity (Lin, 1999; McDonald, 2011). Thus, internships may expand the number and diversity of students' contacts thereby improving employment outcomes for all students, particularly for those students who do not initially have a broad professional network or a network that includes access to "high status" individuals.

Finally, internship experiences may provide symbolic capital via powerful signaling that distinguishes candidates in competitive hiring processes. Since hiring is costly and it is difficult to know in advance how productive an employee will be, employers look for clues or "signals" that may convey information about the quality of candidates (Connelly et al., 2011; Spence, 1973). Internships—particularly those that are competitive to secure—could be such a signal, indicating otherwise hidden but desirable qualities in a hire, including work ethic, motivation, and ability to thrive in a professional environment. There is evidence that employers view internships in this light: in an experiment sending fictionalized résumés to prospective employers, Nunley et al. (2016) demonstrated that candidates with internship experience were more likely to be called for an interview, even when the internship experience is so dated and brief that relevant skill development would be negligible.

The importance of each of these mechanisms has practical application. If the primary mechanism by which internships convey career benefits is via signaling, expanding access to internships would allow more students to possess the signal, broadening the talent pool for prospective employers. At the same time, if highly competitive and paid internships confer more potent signaling, broadening the pool may make the signal less differentiating for prospective employers. If the primary mechanism is the development of skills, both students and employers stand to gain from broader access to internships, as increased productivity should lead to higher wages and higher profits. If the most important mechanism is professional connections, some students may stand to lose and others to gain when access is broadened. Students who already had professional connections, for example via family or friends, would no longer have the same advantages, while students who lacked those connections would have better opportunities than they otherwise would have had. Employers should gain in this scenario as they would have familiarity and connections with a broader talent pool.

Other types of work-based learning beyond internships are already more broadly accessible but may not deliver the same types of career benefits. For example, co-ops and practica are generally unpaid and mandatory in particular disciplines and facilitated by an institution (Hora et al., 2017). This means every student in a particular major will have access, but the signaling mechanism would not apply. Work-study programs are primarily financial aid mechanisms that place students in part-time campus or community jobs; thus, they are accessible to students who are financially disadvantaged, but these students are not typically

placed into roles that develop broadly applicable skills or a professional network (Leonard et al., 2021). This study examines the links between each of these types of work-based learning and a range of career outcomes over time to better understand which experiences are likely to benefit students and why.

Data

To better isolate the relationship between internships and other forms of work-based learning and students' post-completion success, we use data from the Baccalaureate and Beyond longitudinal study. This nationally representative longitudinal study of bachelor's degree graduates, conducted for the National Center for Education Statistics, follows graduates from the class of 2016, who were surveyed again in 2017 and 2020. The cohort is drawn from the National Postsecondary Student Aid Study (NPSAS). NPSAS employs a two-stage sampling design in which first a nationally representative sample of institutions is selected (including public and private, non-profit and for-profit institutions), and then students are sampled from those institution records. Because NPSAS captures both students who did and did not receive student financial aid and students from all institutions, its results generalize to the entire population of U.S. postsecondary students. Drawing from this NPSAS cohort, the Baccalaureate and Beyond cohort follows approximately 26,000 graduating seniors, contacting them one year and four years after graduation. The weighted response rate was 71 percent at the one-year mark and 63 percent at the four-year mark (Wine et al., 2019, 2023). Table 15 in the appendix shows the demographic characteristics of respondents to the survey (Henderson et al., 2022).

The survey captures participation in a broad spectrum of work-based learning models. For this study we use two questions. The first is BB17BINTERN, which reads as follows:

Prior to completing your bachelor's degree in [BA COMPLETION DATE], did you participate in any of the following?

- Paid internship.
- Unpaid internship.
- Co-operative experience.
- Practicum.

1 = Yes 0 = No.

Students could have had more than one of these types of experiences and were able to select as many as applied. In addition, we also look at whether a student received work-study funding. This is variable, TOTWKST is based on student-level financial aid data and is equal to the sum of all federal work-study (TFEDWRK), state work-study (STWKAMT), and institutional work-study (INSTWRK).

This participation varies by race and ethnicity, as shown in Table 1.[1] Black or African American students, Hispanic students, American Indian and Native Hawaiian students are

[1] All estimates of descriptive and inferential statistics in this article use survey weights. However, unweighted sample sizes are also shown in the tables of descriptive statistics so that the reader can see the actual number of observations. These unweighted sample sizes are rounded to the nearest ten to protect privacy per IES guidelines.

Table 1 Participation rates in work-based learning experiences by race/ethnicity

Race/Ethnicity	Paid internship (%)	Unpaid internship (%)	Co-op (%)	Practicum (%)	Work study (%)	Unweighted n
White	30.3	30.0	9.4	15.5	9.5	12,630
Black or African American	20.8	30.4	12.0	18.5	9.8	2610
Hispanic	24.2	32.6	12.1	13.3	10.1	2890
Asian	32.9	32.9	10.7	11.9	10.5	1320
American Indian/Alaska Native	24.9	31.7	9.8	23.2	5.0	110
Native Hawaiian/Other Pacific Islander	12.1	33.4	11.4	14.5	5.3	80
More than one race	31.6	34.1	11.2	11.8	9.7	700
Total	28.7	30.8	10.2	15.2	9.7	20,330

Source: U.S. Department of Education, National Center for Education Statistics, 2016/20 Baccalaureate and Beyond Longitudinal Study (B&B:2016/20). Note: row totals do not sum to 100% as some students had more than one of these experiences and some had none of these experiences

Table 2 Participation rates in work-based learning experiences by gender

Gender	Paid internship (%)	Unpaid internship (%)	Co-op (%)	Practicum (%)	Work study (%)	Unweighted n
Male	34.0	25.3	10.2	10.3	8.9	8110
Female	24.8	34.9	10.2	18.8	10.3	12,220
Total	28.7	30.8	10.2	15.2	9.7	20,330

Source: U.S. Department of Education, National Center for Education Statistics, 2016/20 Baccalaureate and Beyond Longitudinal Study (B&B:2016/20). Note: row totals do not sum to 100% as some students had more than one of these experiences and some had none of these experiences.

less likely to participate in paid internships compared to other groups, but are more likely to participate in unpaid internships. Fewer students across the board participate in co-ops, practica, and work study programs. We also see differences by gender, with male students more likely than female students to participate in paid internships, and female students more likely to participate in unpaid internships. Female students are also more likely than their male counterparts to participate in practica, as shown in Table 2.

Participation in paid internships also varies by field of study, as shown in Table 3. More than half of engineering students reported participating in a paid internship, while only about one in ten students in healthcare, general studies fields, and education reported completing a paid internship. Education and healthcare majors are most likely to report participating in practicum experiences—this is unsurprising, as a practica is often required in these fields of study. Computer science and business majors are more likely to have participated in a paid internship compared to any other type of work-based learning, while majors in other applied fields, humanities, social sciences, and general studies are most likely to participate in an unpaid internship. Biological and physical science majors are about evenly split between participation in paid versus unpaid internships.

Other student characteristics related to differences in work-based learning participation include institutional selectivity, family income quartile, parental education, student GPA, and student test scores. As expected, students from more selective institutions, higher

Table 3 Participation rates in work-based learning experiences by field of study

Field of study	Paid internship (%)	Unpaid internship (%)	Co-op (%)	Practicum (%)	Work study (%)	Unweighted n
Engineering	55.7	15.0	14.8	4.2	10.2	1190
Computer and information sciences	41.0	13.6	8.3	3.5	7.7	1500
Business	39.1	23.3	7.6	7.5	6.5	2580
Biological and physical sciences	32.6	30.5	10.1	8.0	15.1	1910
Other applied fields	28.0	47.2	11.3	17.0	8.3	3310
Humanities	23.0	35.0	9.5	12.5	15.3	1510
Social sciences	22.4	35.8	8.0	12.8	10.7	2130
Health care fields	13.9	26.0	9.8	31.0	6.2	3000
General studies and other	10.0	23.0	5.3	13.6	5.4	290
Education	9.5	35.8	24.7	59.1	10.4	2080
Total	28.7	30.8	10.2	15.2	9.7	19,490

Source: U.S. Department of Education, National Center for Education Statistics, 2016/20 Baccalaureate and Beyond Longitudinal Study (B&B:2016/20). Note: row totals do not sum to 100% as some students had more than one of these experiences and some had none of these experiences.

income families, more educated families, and those with higher GPA and test scores are more likely to participate in paid internships. (See Appendix for Table 16, 17, 18).

Method of Analysis

Since these demographic and academic characteristics of a student are associated with their likelihood to access certain types of work-based learning and may be related to their post-completion outcomes, in order to estimate the relationship between work-based learning, including paid and unpaid internships, practica, co-ops, and work-study, and career outcomes we need a way to address selection effects—for this we use a matching design to more closely approximate experimental conditions. The following paragraphs describe the control variables chosen, the technique used—nearest neighbor matching followed by regression—and the basis for this choice.

We identified relevant control variables based upon a growing body of research that shows that students' access to work-based learning experiences is shaped by multiple social and academic factors, including students' socioeconomic status, racial and gender identity, and field of study (e.g., Frenette, 2015; Jang-Tucci et al., 2025; Torpey-Saboe et al., 2023). These influences rarely operate in isolation. Instead, they combine to produce differences across student groups. To isolate our ability to examine relationships among work-based learning participation and post-completion earnings, it is important to control for background and academic characteristics that have consistently been linked to labor market earnings. For example, race and gender shape employment opportunities and pay gaps such that women and persons of color in the U.S. have lower earnings compare to their male and White counterparts (Hu & Wolniak, 2013; Olitsky, 2014). Similarly, academic preparation and performance, including GPA and test scores, predict college completion and early-career earnings such that those with higher GPAs and test scores have higher earnings (Hu & Wolniak, 2013). Field of study is one of the strongest predictors of post-college wages, with large discrepancies across STEM, business, health, and arts fields (Altonji & Zhong,

2021). Institutional selectivity also has been shown to influence earnings through employer recruitment patterns and network advantages such that higher institutional prestige and network connections confer greater benefits (Witteveen & Attewell, 2017). Finally, parental education and household income shape students' educational pathways and labor-market trajectories such that students from lower income households have more restricted pathways (Delaney et al., 2011). Controlling for these factors helps distinguish the specific contribution of work-based learning experiences from pre-existing factors that already affect economic outcomes.

First, we examine the distribution of the covariates among both the potential treatment and control groups (those students who did and did not have a particular type of work-based learning experience, such as a paid internship). This first step is necessary to establish that there is a common support—that is, there is sufficient overlap in the covariates of the treatment and control group such that each treated observation can be reasonably matched to a similar control observation. An illustrative example showing common support with a single covariate—high school SAT scores—is shown below.

As Fig. 1 illustrates, while the distribution of SAT scores is different among students who have a paid internship and students who do not, there are students from both groups in every SAT range, allowing for matching on this characteristic. There is also overlap for the treatment and control groups across each category of high school and college GPA, field of study, gender, and race/ethnicity. While the comparison of the distribution of all the covariates across the potential treatment and control groups cannot be shown here due to limitations related to disclosure review processes, the distribution of the propensity scores across potential treatment and control groups shows that there is enough common support to avoid the necessity of trimming observations.

Most estimates using observational data are model-dependent. Whether or not a statistically significant relationship is detected can change based on the variables included in

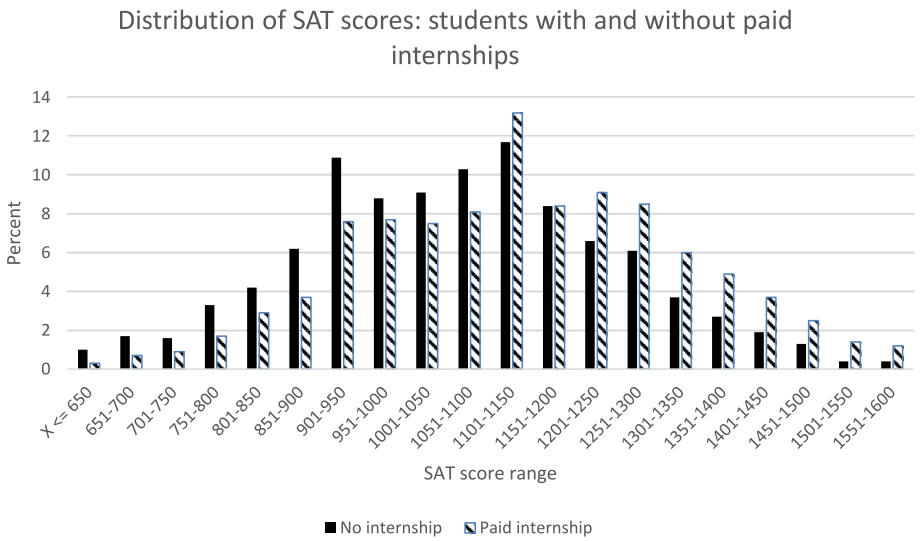


Fig. 1 Distribution of SAT scores among students with and without paid internships (raw data). *Source*: U.S. Department of Education, National Center for Education Statistics, Baccalaureate and Beyond: 2016/2020 (B&B). Computation by NCES PowerStats on 11/3/2025 at 11:26:18

the model, the chosen functional form, the transformation of the dependent variable, and whether a specific subset of the sample is analyzed. This is especially true when one or more control variables are related to (collinear with) the treatment variable. Breaking or weakening the link between the control variables and the treatment variable by pre-processing the data via matching allows us to adjust for control variables without making parametric assumptions. This technique greatly reduces the potential for bias and makes estimates more stable across different models (Ho et al., 2007). We chose Mahalanobis distance matching (MDM), rather than propensity score matching (PSM), as PSM can be more vulnerable to model dependence and bias (King & Nielsen, 2019). MDM has the advantage of approximating a fully blocked randomized experiment by matching on all covariates, rather than just the predicted likelihood—or propensity—of an observation to fall in either the treatment or control groups based on a logistic regression model, as in PSM.

To estimate the relationship between paid internships and other forms of work-based learning and earnings and employment outcomes, we first use nearest neighbor matching to pre- process the data to approximate experimental conditions.[2] Students in the treatment group (those who held a paid internship) are matched with students in the control group (those who did not hold a paid internship) according to several factors described above that are shown to be related to whether a student was likely to have a paid internship (the treatment variable). These include field of study, selectivity of college, college GPA, high school GPA, sat score, family income, parental education, gender, race and ethnicity.[3] Students who are part of the control group and are not matched to a similar student who held a paid internship were dropped from the analysis.

Table 4 shows a comparison of the difference in the means between the raw and the matched data, as well as the variance ratio of the raw and matched data. The standardized differences show the difference between the means of each variable in the treatment group (those who had a paid internship) and the control group (those who did not have a paid internship). The raw column is the difference before matching on all the demographic and academic covariates listed above, while the matched column shows the difference after matching. Similarly, the variance ratio columns are the ratio of the variance for the treated observations (those who had a paid internship) and the control (those who did not). Ideally the difference in means for the matched data should be 0, or very close to it, while the variance ratio should be as close as possible to 1 (meaning the variances are close to equal). This is because matching aims to create conditions that are as close as possible to the experimental ideal, where the only difference between the treatment and control groups, on average, is the treatment variable, meaning that the distribution of all other variables should be approximately equal between the treatment and control groups. The table demonstrates that this covariate balance is improved for all variables after matching.

A simple difference in means test—a commonly used technique for estimating the impact of a treatment variable after matching—assumes that the covariates are no longer related to the outcome variable. This will only be the case when exact matching on every covariate is

[2] Nearest-neighbor matching is implemented using Stata's teffects nnmatch command to estimate the average treatment effect (ATE). This procedure performs one-to-one Mahalanobis distance matching with replacement, matching treated and control observations in both directions.

[3] Parents' highest education and institutional selectivity are modeled as ordered variables reflecting increasing levels of educational attainment and selectivity. Although these measures are categorical, treating them as ordered captures their underlying structure while limiting the number of parameters in matching and regression models. Alternative specifications using categorical indicators yielded substantively similar results.

Table 4 Covariate balance summary: paid internships (subsample analysis)

	Standardized differences		Variance ratio	
	Raw	Matched	Raw	Matched
Black	−0.222	−0.017	0.567	0.961
Hispanic	−0.117	−0.022	0.761	0.952
Female	−0.296	0.009	1.012	0.999
Parental education	0.262	0.088	0.852	0.856
Income quartile	0.230	0.075	1.015	0.958
Computer science	0.062	0.000	1.202	1.000
Engineering	0.421	0.000	2.854	1.000
Biological and physical sciences	0.047	0.000	1.104	1.000
General studies and other	−0.109	0.000	0.346	1.000
Social sciences	−0.190	0.000	0.626	1.000
Humanities	−0.149	0.000	0.618	1.000
Health care fields	−0.364	−0.001	0.323	0.999
Business	0.217	0.001	1.506	1.001
Education	−0.130	0.000	0.515	1.000
Other applied	−0.026	0.000	0.951	1.000
Bachelor's GPA	0.194	0.082	0.899	0.881
High school GPA	0.332	0.122	0.715	0.779
SAT scores	0.505	0.120	1.017	0.826
Institutional selectivity	−0.459	−0.133	0.795	0.857

Sample characteristics

Number of observations (unweighted) = 10,140

(Matched) Treated observations = 5070

(Matched) Control observations = 5070

Source: U.S. Department of Education, National Center for Education Statistics, 2016/20 Baccalaureate and Beyond Longitudinal Study (B&B:2016/20).

possible, which is not the case with this data. Therefore, to estimate the relationship between a paid internship and post-completion earnings, we run a linear regression, using all the matching variables as covariates to avoid omitted variable bias (Ho et al., 2007).

We perform the matching separately for the five different work-based learning variables: paid internships, unpaid internships, practica, co-op, and work-study. Each time, we restrict the sample to exclude students who had a different type of work-based learning in addition to the one of interest. For example, in our analysis of paid internships, we exclude anyone who had an unpaid internship, practicum, co-op, or work-study experience so that the comparison of the treatment and the control groups is between someone who had only a paid internship and someone who had none of these experiences. Similarly, the analysis of unpaid internships compares those who had only an unpaid internship to those who had none of the five experiences. As a result, estimates provide population-representative estimates for the subpopulation of undergraduates with a single work-based learning experience of the type under study, rather than for all undergraduates. Table 22 in the appendix shows the respective sample sizes for each of these subpopulations. Tables 5, 6, 7 and 8 compare the raw and matched means and variance ratios for unpaid internships, practica, co-op, and work-study experiences.

We then run four different regression models on each of the matched datasets. In each case, the covariates remain the same, but the outcome variable changes. The four different outcome variables we look at are: (1) annual earnings 12 months after graduation; (2) annual earnings four years after graduation; (3) respondent's perception of whether a bachelor's degree was

Table 5 Covariate balance summary: unpaid internships (subsample)

	Standardized differences		Variance ratio	
	Raw	Matched	Raw	Matched
Black	−0.073	−0.012	0.855	0.976
Hispanic	0.038	−0.010	1.080	0.979
Female	0.193	0.009	0.902	0.996
Parental education	0.030	0.031	0.987	0.860
Income quartile	0.023	0.006	1.066	1.013
Computer science	−0.231	0.000	0.361	1.000
Engineering	−0.157	0.000	0.477	1.000
Biological and physical sciences	−0.039	0.000	0.915	1.000
General studies and other	−0.031	0.000	0.788	1.000
Social sciences	0.009	-0.001	1.019	0.999
Humanities	0.073	0.000	1.203	1.000
Health care fields	−0.132	0.000	0.740	1.000
Business	−0.109	0.000	0.753	1.000
Education	0.114	0.000	1.517	1.000
Other applied	0.302	0.000	1.534	1.001
Bachelor's GPA	0.112	0.033	0.934	0.888
High school GPA	0.122	0.072	0.905	0.849
SAT scores	0.050	0.035	1.024	0.910
Institutional selectivity	−0.168	−0.066	1.045	0.979

Sample characteristics

Number of observations (unweighted) = 10,620 (Matched)

(Matched) Treated observations = 5310

Control observations = 5310

All fields except "Social sciences" and "Other applied" show exact matching

Source: U.S. Department of Education, National Center for Education Statistics, 2016/20 Baccalaureate and Beyond Longitudinal Study (B&B:2016/20).

required by their employer for them to be hired for their first job after graduation[4] and (4) respondent's perception that their first job was part of a career.[5] For example, for paid internships, the models were as follows[6]:

(1) $\textit{1st Year Annual Earnings} = \alpha + \beta_1 \cdot \text{Paid internship} + \beta_2 \cdot \text{Black} + \beta_3 \cdot \text{Hispanic} + \beta_4 \cdot \text{Female} + \beta_5 \cdot \text{Parental Education} + \beta_6 \cdot \text{Income quartile} + \beta_{7\text{–}15} \cdot \text{Field of study dummy variables} + \beta_{16} \cdot \text{Bachelor's GPA} + \beta_{17} \cdot \text{High school GPA} + \beta_{18} \cdot \text{SAT scores} + \beta_{19} \cdot \text{Institutional selectivity} + \epsilon.$

(2) $\textit{4th Year Annual Earnings} = \alpha + \beta_1 \cdot \text{Paid internship} + \beta_2 \cdot \text{Black} + \beta_3 \cdot \text{Hispanic} + \beta_4 \cdot \text{Female} + \beta_5 \cdot \text{Parental Education} + \beta_6 \cdot \text{Income quartile} + \beta_{7\text{–}15} \cdot \text{Field of study dummy variables} + \beta_{16} \cdot \text{Bachelor's GPA} + \beta_{17} \cdot \text{High school GPA} + \beta_{18} \cdot \text{SAT scores} + \beta_{19} \cdot \text{Institutional selectivity} + \epsilon$

[4] The exact question wording is "Was a bachelor's degree required by [EMPLOYER NAME] for you to be hired as a(n) [JOB TITLE]?".

[5] Exact question wording: "Did you consider your job as a(n) [JOB TITLE] with [EMPLOYER NAME] to be part of a career you were pursuing?".

[6] Nearest-neighbor matching is used as a preprocessing step to restrict analyses to observations on common support and is conducted without survey weights. Outcome models are then estimated using survey-weighted regressions with analytic weights (wta000) to account for unequal probabilities of selection. Standard errors are computed using Stata's svy linearization method. Although the restricted-use Baccalaureate and Beyond data include design variables that permit clustering at the institutional level, because paid internship participation and post-baccalaureate earnings vary within institutions and our estimate is an individual-level treatment effect, we estimate survey-weighted models with linearized standard errors and do not impose additional institutional clustering.

Table 6 Covariate balance summary: practica (subsample)

	Standardized differences		Variance ratio	
	Raw	Matched	Raw	Matched
Black	−0.150	−0.064	0.703	0.865
Hispanic	−0.067	−0.061	0.862	0.872
Female	0.415	0.020	0.724	0.990
Parental education	0.052	0.093	0.903	0.805
Income quartile	0.048	0.034	0.999	0.947
Computer science	−0.353	−0.002	0.104	0.993
Engineering	−0.290	0.000	0.134	1.000
Biological sciences	−0.342	0.000	0.316	1.000
General studies	−0.057	0.000	0.626	1.000
Social sciences	−0.280	0.000	0.461	1.000
Humanities	−0.182	0.000	0.540	1.000
Health care fields	0.361	0.001	1.655	1.001
Business	-0.351	−0.001	0.272	0.998
Education	0.971	0.001	5.224	1.002
Other applied	−0.227	0.001	0.576	1.001
Bachelor's GPA	0.368	0.063	0.802	0.837
High school GPA	0.196	0.122	0.822	0.847
SAT scores	0.073	0.079	0.805	0.777
Institutional selectivity	−0.110	−0.195	0.747	0.676

Sample characteristics

Number of observations (unweighted) = 4410

(Matched) Treated observations = 4410

(Matched) Control observations = 4410

Source: U.S. Department of Education, National Center for Education Statistics, 2016/20 Baccalaureate and Beyond Longitudinal Study (B&B:2016/20).

(3) $Probability\ job\ requires\ a\ degree = \frac{e^{(\beta 0+\beta 1-19\cdot \mathrm{X})}}{1+e^{(\beta 0+\beta 1-19\cdot \mathrm{X})}}$

(4) $Probability\ job\ considered\ part\ of\ career = \frac{e^{(\beta 0+\beta 1-19\cdot \mathrm{X})}}{1+e^{(\beta 0+\beta 1-19\cdot \mathrm{X})}}$

Results

We find that paid internships are linked with an earnings bump of $5564 one year after graduation after matching and then controlling for the various academic and demographic covariates (see Table 9). Given that the average earnings for all students one year after graduation is $36,379, the earnings boost associated with a paid internship is equal to 15% of the average annual salary. None of the other work-based learning experiences are significantly related to first-year annual earnings (see Tables 10 through 13). Many of the covariates in the models are still significant, demonstrating that regression, as opposed to a simple t-test, is the appropriate approach.

The earnings difference for paid internships does not fade, but rather appears to increase over time, as shown in Table 9. Matching on the same covariates yields an estimated boost in earnings of $9,089 per year for students four years after graduation (or 16% of average annual pay). At four years out, unpaid internships have a negative relationship to earnings, and the other experiences—co-ops, practica, and federal work-study—have no significant relationship with earnings (see Tables 10, 11, 12, 13).

Paid internships also predict the likelihood that a graduate's first job will require their degree. Students who had a paid internship are up to 14 percentage points more likely to

Table 7 Covariate balance summary: co-op (subsample)

	Standardized differences		Variance ratio	
	Raw	Matched	Raw	Matched
Black	−0.040	−0.091	0.922	0.817
Hispanic	−0.068	−0.087	0.862	0.820
Female	−0.217	−0.048	1.029	1.012
Parental education	0.001	0.075	0.915	0.697
Income quartile	−0.044	0.007	1.044	0.860
Computer science	−0.052	0.000	0.843	1.000
Engineering	0.187	0.000	1.786	1.000
Biological sciences	−0.017	0.001	0.965	1.002
General studies	−0.061	0.000	0.601	1.000
Social sciences	−0.070	0.001	0.862	1.003
Humanities	0.067	0.000	1.191	1.000
Health care fields	−0.232	−0.001	0.554	0.997
Business	−0.104	−0.001	0.766	0.998
Education	0.261	0.000	2.294	1.000
Other applied	0.026	0.000	1.052	1.000
Bachelor's GPA	0.060	0.022	0.896	0.757
High school GPA	0.052	0.135	0.869	0.817
SAT scores	0.058	0.026	0.916	0.757
Institutional selectivity	−0.035	−0.107	0.981	0.768

Sample characteristics

Number of observations (unweighted) = 7780

(Matched) Treated observations = 3890

(Matched) Control observations = 3890

Source: U.S. Department of Education, National Center for Education Statistics, 2016/20 Baccalaureate and Beyond Longitudinal Study (B&B:2016/20).

report having a first job that requires their degree.[7] Students with an unpaid internship are also slightly more likely to have a first job that requires their degree: up to six percentage points. Co-ops, practica, and federal work-study participation are not linked to differences in perception that the first job after graduation requires a college degree.

The results are similarly strong when it comes to having a first job that is part of a career: graduates who held a paid internship are up to 13 percentage points more likely to report having a first job that is part of a career. Graduates who held a practicum are also more likely to have a first job that is part of a career: up to twelve percentage points. Unpaid internships, co-ops, and federal work-study are not significantly related to differences in perceptions that a graduate's first job is part of a career.

The next question is *why* does a paid internship predict higher earnings even after matching and then controlling on demographic and academic characteristics? As described previously, internships could boost the quality of a graduate's first job either by:

1. Human capital: developing skills that can observed by the employer (which would be the case if the internship- sponsoring employer chooses to hire the candidate on a permanent basis, or if the skills are demonstrated in a performance task or other means during the interview process),
2. Social capital: facilitating a connection or recommendation to an employer, or

[7] Coefficients are log-odds from survey-weighted logistic regression models. To aid interpretation, coefficients are translated in the text into approximate percentage-point changes in outcome probabilities using the divide-by-four rule, which provides the maximum marginal effect of a logit coefficient on the probability scale (Gelman & Hill, 2007). These values represent absolute changes in probability (percentage points), not percent changes. Actual marginal effects vary with baseline probability; the divide-by-four rule provides an upper-bound approximation.

Table 8 Covariate balance summary: work-study (subsample)

	Standardized differences		Variance ratio	
	Raw	Matched	Raw	Matched
Black	−0.107	−0.104	0.788	0.790
Hispanic	−0.076	−0.055	0.846	0.885
Female	0.083	0.014	0.968	0.995
Parental education	0.106	0.127	0.813	0.670
Income quartile	−0.254	−0.093	0.932	0.940
Computer science	−0.107	0.000	0.681	1.000
Engineering	−0.094	0.000	0.670	1.000
Biological sciences	0.380	0.001	1.785	1.003
General studies	−0.154	0.000	0.145	1.000
Social sciences	0.045	0.001	1.094	1.003
Humanities	0.186	0.000	1.530	1.000
Health care fields	−0.234	−0.001	0.549	0.997
Business	−0.139	−0.002	0.689	0.996
Education	0.022	0.000	1.095	1.000
Other applied	−0.174	0.000	0.673	1.000
Bachelor's GPA	0.203	0.105	0.830	0.734
High school GPA	0.429	0.277	0.726	0.685
SAT scores	0.431	0.189	1.006	0.806
Institutional selectivity	−0.598	−0.284	0.631	0.688

Sample characteristics

Number of observations (unweighted) = 8000

(Matched) Treated observations = 4000

(Matched) Control observations = 4000

Source: U.S. Department of Education, National Center for Education Statistics, 2016/20 Baccalaureate and Beyond Longitudinal Study (B&B:2016/20).

3. Symbolic capital: by adding a signal to a resume that sets an intern apart from other applicants.

This can be partially tested via the dataset, which included a series of questions on how a graduate secured their first job, including whether it was through an internship. It also asks whether their first job was related to their internship. Table 14 shows the relationship we might expect between these variables and job quality measures such as earnings, whether a job requires a degree, and whether a job is part of a career were each of our hypotheses to be true.

If internships are related to better job quality due to skill development, then we might expect positive signs on the coefficients for securing a job through an internship and having a job related to the internship, and for these variables to explain most of the variation, leaving little or nothing to be explained by the generic variable on completing a paid internship. A candidate who developed and demonstrated skills on the job would be more likely to get a job through that internship or to demonstrate those skills to get a job related to their internship work. Apart from these channels to demonstrate skill development, internships would not be valuable.

If the primary mechanism is social capital, we might expect to see a positive sign on the variable for securing a job through an internship, but not necessarily on a job that is related to the internship, as it's the personal connections rather than the skill overlap that matter. Finally, if the mechanism is symbolic capital—simply a screening device for employers—we would expect to see no relationship or relatively weak relationships between securing a job through an internship or the relationship between a job and an internship and job quality, with the bulk of the variation remaining in the general variable for paid internship participa-

Table 9 Paid internships and early career outcomes

Outcome variable	1st year annual earnings	4th year annual earnings	Job requires degree	Career-related job
Paid internship	5563.52**	9089.45**	0.548**	0.522**
	(850.87)	(1500.52)	(0.092)	(0.099)
Black	−327.42	−4494.99**	−0.167	−0.331*
	(1190.20)	(1725.79)	(0.156)	(0.139)
Hispanic	−1718.11	−353.27	0.031	−0.063
	(1003.89)	(2158.37)	(0.143)	(0.132)
Female	−2501.97**	−6838.72**	−0.232*	−0.082
	(745.77)	(1395.77)	(0.094)	(0.090)
Parental education	34.2	97.93	0.053*	−0.001
	(186.30)	(372.74)	(0.024)	(0.023)
Income quartile	1339.07**	1684.18**	0.055	0.097*
	(326.68)	(615.51)	(0.041)	(0.040)
Field of study (ref: Computer science)				
Engineering & engineering technology	−3109.93	−7499.63	0.275	−0.492*
	(2521.11)	(4884.35)	(0.209)	(0.236)
Biological & physical sciences	−21,388.31**	−31,518.90**	−0.565**	−0.705**
	(2409.09)	(4675.42)	(0.204)	(0.222)
General studies & other	−20,497.44**	−23,194.79**	−0.688	−0.697
	(3402.15)	(6226.87)	(0.409)	(0.358)
Social sciences	−19,763.25**	−28,411.75**	−0.451*	−0.904**
	(2310.32)	(4682.12)	(0.210)	(0.222)
Humanities	−25,307.68**	−36,126.94**	−0.545*	−0.924**
	(2371.98)	(4773.48)	(0.225)	(0.233)
Health care fields	−8670.62**	−15,137.43**	−0.257	−0.173
	(2497.61)	(4797.32)	(0.226)	(0.245)
Business	−13,216.82**	−16,759.94**	−0.223	−0.677**
	(2320.99)	(4746.29)	(0.197)	(0.217)
Education	−21,213.03**	−32,410.48**	0.147	−0.271
	(2576.49)	(4714.15)	(0.285)	(0.325)
Other applied	−20,975.87**	−25,997.67**	−0.375	−0.715**
	(2320.95)	(4703.04)	(0.207)	(0.224)
Bachelor's GPA	1350.01	831.14	0.416**	0.217*
	(866.43)	(1600.72)	(0.104)	(0.102)
High school GPA	−561.23	−1273.94	0.015	0.029
	(439.01)	(694.43)	(0.051)	(0.047)
SAT scores	2.19	7.32	−0.00003	0.0003
	(2.30)	(4.80)	(0.00027)	(0.0003)
Institutional selectivity	−832.34	−5606.98**	−0.288**	−0.048
	(603.25)	(1023.51)	(0.064)	(0.058)
Constant	44,449.93**	85,938.26**	−1.385**	−0.347
	(4584.69)	(8829.90)	(0.524)	(0.503)
Observations (unweighted)	5070	4140	5070	5070

Table 9 (continued)

Outcome variable	1st year annual earnings	4th year annual earnings	Job requires degree	Career-related job
Prob > F	0	0	0	0
R-squared	0.196	0.184	–	–

* $P \leq 0.05$, ** $P \leq 0.001$

Standard errors in parentheses.

Source: U.S. Department of Education, National Center for Education Statistics, 2016/20 Baccalaureate and Beyond Longitudinal Study (B&B:2016/20).

tion, as this hypothesis treats the value of internships as interchangeable and generic regardless of skill overlap or personal connections.

When including these variables in the regression model, we find that securing a first job through an internship boosts annual earnings one year after graduation by $3452—on top of the estimated $4850 earnings boost for having had a paid internship (see Table 15). Having a first job that is related to one's internship also appears to carry earnings benefits, with a boost of $1274 one year after graduation. Finding a first job through an internship also boosts the probability of having a first job that requires a college degree (+12 percentage points) and that the job is part of a career (+14 percentage points), while having a job related to an internship increases the probability of a college level job by up to 4 percentage points and the probability of a job that is part of a career by up to 6 percentage points. Given that all three variables remain positively and significantly related to the measures of job quality, there is some support for all three hypotheses: human capital, social capital, and symbolic capital. Nevertheless, the support for the human capital hypothesis is the weakest, as the magnitude of the coefficient on that variable (first job related to internship) is only a fraction of that for the other variables. The evidence for the social capital and symbolic capital mechanisms is somewhat stronger. We note that job search and hiring processes are sequential, such that signaling and access to networks may influence earlier stages (such as interviews and initial job offers), whereas human capital gains may manifest later through job performance, advancement, or earnings growth. As a result, the outcomes examined here may be more sensitive to signaling and social capital mechanisms than to longer-term human capital accumulation.

To further explore the relationships among internships, finding a job via an internship, and post-completion earnings, we conduct mediation analysis. A mediating variable can be thought of as the pathway through which the independent variable is linked to the dependent variable—the *why* or the *how* (William, 2024). Mediation analysis works by:

1. Regressing Y on X to get the total association- in this case, the relationship between a paid internship and annual earnings.
2. Regressing M (the mediator) on X to see, in this case, whether having had a paid internship is linked to whether someone finds a job through an internship.
3. Regressing Y on X + M to understand the direct association (the coefficient on X when M is included in the regression) and the indirect association (the difference between the total association and the direct association). In our example, this provides an estimate of how much of the earnings boost linked to a paid internship (the total association) comes via finding a job through an internship (the indirect association).

Table 10 Unpaid internships and early career outcomes

Variable	1st year annual earnings	4th year annual earnings	Job requires degree	Career-related job
Unpaid internship	−1280.43	−2350.75*	0.25**	−0.04
	(656.76)	(1193.83)	(0.09)	(0.09)
Black	−864.44	−5136.97**	−0.16	−0.29*
	(1050.92)	(1493.13)	(0.15)	(0.13)
Hispanic	165.36	−491.94	0.09	−0.04
	(906.85)	(1719.23)	(0.13)	(0.12)
Female	−2186.65**	−6134.81**	−0.17	−0.07
	(693.59)	(1235.69)	(0.09)	(0.09)
Parental education	−121.91	13.48	0.04	0.02
	(162.29)	(334.74)	(0.02)	(0.02)
Income quartile	1572.34**	1511.4**	0.06	0.05
	(286.54)	(542.23)	(0.04)	(0.04)
Field of study (ref: Computer Science)				
Engineering & engineering technology	−1186.48	−4150.85	0.17	−0.74**
	(3023.83)	(4190.78)	(0.27)	(0.28)
Biological & physical sciences	−16,970.58**	−23,623.55**	−0.70**	−0.69**
	(2487.20)	(3749.75)	(0.24)	(0.24)
General studies & other	−14,940.36**	−18,480.14**	−0.76*	−0.69*
	(3305.50)	(5303.68)	(0.37)	(0.34)
Social sciences	−16,861.98**	−25,501.33**	−0.53*	−0.91**
	(2432.10)	(3613.33)	(0.24)	(0.24)
Humanities	−19,116.21**	−27,060.44**	−0.71**	−0.95**
	(2501.09)	(3811.02)	(0.25)	(0.25)
Health care fields	−4587.46	−10,568.76**	−0.41	−0.05
	(2601.74)	(3827.68)	(0.25)	(0.25)
Business	−10,362.20**	−11,778.01**	−0.55*	−0.82**
	(2498.45)	(3826.67)	(0.24)	(0.24)
Education	−13,577.11**	−23,521.68**	0.32	−0.21
	(2766.09)	(3791.54)	(0.29)	(0.31)
Other applied	−15,704.59**	−19,584.53**	−0.56*	−0.78**
	(2413.43)	(3672.43)	(0.23)	(0.23)
Bachelor's GPA	−29.38	1371.44	0.28**	0.18
	(799.80)	(1355.63)	(0.10)	(0.10)
High school GPA	−886.97*	−1250.86*	−0.03	0.04
	(373.88)	(631.26)	(0.05)	(0.04)
SAT scores	−0.79	2.57	0	0
	(2.10)	(3.59)	(0.00)	(0.00)
Institutional selectivity	−257.11	−1968.59*	−0.24**	0.08
	(422.84)	(886.86)	(0.06)	(0.05)
Constant	48,572.51**	76,563.41**	−0.83	−0.63
	(4123.65)	(7170.68)	(0.51)	(0.48)
Observations (unweighted)	5310	4330	5310	5310

Table 10 (continued)

Variable	1st year annual earnings	4th year annual earnings	Job requires degree	Career-related job
Prob > F	0	0	0	0
R-squared	0.12	0.11	–	–

* $P \leq 0.05$, ** $P \leq 0.001$

Source: U.S. Department of Education, National Center for Education Statistics, 2016/20 Baccalaureate and Beyond Longitudinal Study (B&B:2016/20).

The standard errors reported from this mediation analysis are calculated using the delta method, which accounts for uncertainty from both the outcome and mediator models in the matched analytic sample. These standard errors are suitable for linear, logistic, and other regression-based mediation pathways and allow valid inference for estimates of direct, indirect, and total associations within the matched sample.

When conducting mediation analysis, it appears that the direct association of a paid internship on earnings is much stronger than the indirect association it has via a graduate finding their first job through an internship (see Table 16, 17). Out of a total association of $5325 higher first-year annual earnings in this model, the direct association of a paid internship is $4142 (78 percent), while the indirect association of finding the job through the internship is only $1183 (22 percent). Similarly, for the likelihood of securing a first job that requires a degree, 87 percent of the total association of a paid internship is direct, while only 13 percent is indirect. These results bolster the case for the symbolic capital hypothesis.

Discussion and Conclusion

Paid internships are associated with higher earnings both immediately after graduation and four years after completion and are also linked to higher likelihood of securing a job that requires a college degree and a job that is part of a career, which aligns with the previous literature (e.g., Bolli et al., 2021). This finding is after matching and controlling for demographic characteristics and academic characteristics of students and institutions. Furthermore, regression analysis and mediation analysis demonstrate that finding a job through an internship or having a first job that is related to an internship both contribute to higher earnings, but that most of the association between a paid internship and earnings is direct and not via these indirect pathways. Other work-based learning experiences had no significant link to post-completion earnings, though unpaid internships were linked to higher likelihood of securing a job that requires a degree and practicum experiences were related to higher likelihood of being in a job that a student perceived to be part of a career.

We hypothesized three key mechanisms by which work-based learning might improve employment outcomes: development of transferable skills, networking that facilitates access to professional connections, and resume signaling for employers that demonstrates otherwise unobservable qualities of a candidate. Our analysis provides some support to all three mechanisms, but the strongest case for the third mechanism: symbolic capital/ resume signaling.

When considering the other types of work-based learning experiences outside of paid internships–unpaid internships, co-ops, practica, and work-study—there are several reasons

Table 11 Practica and early career outcomes

Variable	1st year annual earnings	4th year annual earnings	Job requires degree	Career-related job
Practica	2075.55	1836.19	0.22	0.47**
	(1125.49)	(2919.77)	(0.15)	(0.15)
Black	−588.61	−5463.07**	−0.15	−0.28
	(1183.98)	(1756.15)	(0.16)	(0.14)
Hispanic	−993.66	−882.52	−0.03	−0.06
	(995.66)	(2136.38)	(0.16)	(0.14)
Female	−2151.01**	−7644.88**	−0.27*	−0.02
	(785.32)	(1525.24)	(0.11)	(0.10)
Parental education	−154.94	−147.34	0.04	0
	(181.58)	(379.90)	(0.03)	(0.03)
Income quartile	1387.02**	1588.69**	0.06	0.09*
	(326.04)	(602.24)	(0.05)	(0.04)
Field of study (ref: Computer Science)				
Engineering & engineering technology	−3842.94	−2239.48	0.37	−0.83**
	(3206.64)	(4435.05)	(0.30)	(0.30)
Biological & physical sciences	−16,893.26**	−19,824.74**	−0.41	−0.63*
	(2753.76)	(3876.37)	(0.27)	(0.26)
General studies & other	−17,529.11**	−17,656.53**	−0.75	−0.80*
	(3651.43)	(5524.09)	(0.47)	(0.37)
Social sciences	−17,126.25**	−20,817.00**	−0.12	−0.95**
	(2679.89)	(4004.51)	(0.27)	(0.26)
Humanities	−20,982.72**	−24,201.19**	−0.43	−0.95**
	(2760.74)	(4049.63)	(0.29)	(0.27)
Health care fields	−3825.44	−4162.94	−0.16	−0.02
	(2804.91)	(3977.99)	(0.28)	(0.27)
Business	−10,147.76**	−8745.95*	−0.25	−0.89**
	(2752.82)	(4058.13)	(0.27)	(0.26)
Education	−16,667.06**	−22,096.57**	0.34	−0.31
	(2821.94)	(3975.49)	(0.29)	(0.30)
Other applied	−17,512.73**	−18,508.35**	−0.24	−0.72**
	(2667.40)	(3946.48)	(0.27)	(0.26)
Bachelor's GPA	893.84	391.06	0.33**	0.11
	(859.25)	(1808.95)	(0.12)	(0.11)
High school GPA	−804.24	−1397.85*	−0.01	0.01
	(414.63)	(710.93)	(0.05)	(0.05)
SAT scores	−1.43	0.66	−0.00	0
	(2.31)	(4.16)	(0.00)	(0.00)
Institutional selectivity	−613.55	−2381.95*	−0.22**	0.02
	(487.34)	(998.57)	(0.07)	(0.06)
Constant	47,850.39**	83,313.74**	−0.81	0.16
	(4616.31)	(8635.16)	(0.59)	(0.55)
Observations (unweighted)	4410	3610	4410	4410

Table 11 (continued)

Variable	1st year annual earnings	4th year annual earnings	Job requires degree	Career-related job
Prob > F	0	0	0	0
R-squared	0.14	0.11	–	–

* $P \leq 0.05$, ** $P \leq 0.001$

Source: U.S. Department of Education, National Center for Education Statistics, 2016/20 Baccalaureate and Beyond Longitudinal Study (B&B:2016/20).

we may not observe an earnings boost for participants. Differences in the underlying nature of these experiences may fundamentally change the likelihood that students gain professional connections, skills, or a powerful signal to employers. For example, work-study jobs that are primarily on-campus, not related to field of study, and are awarded based on financial need rather than other employability characteristics may not create professional connections off campus, develop field relevant skills, or signal differentiating quality to employers. Practicum experiences likely help students develop relevant skills and connections but are unlikely to be a differentiating signal to employers as they are often mandatory for certain fields of study.

Unpaid internships could be less likely to be linked with higher wages for several reasons. Unpaid internships may be more likely to be offered via employers that also pay their entry-level workers lower salaries (Hora et al., 2017). Therefore, any professional connections made would be more likely to lead to similarly low-paid work. In addition, unpaid interns may be doing lower-level work and not developing the same level of skills as their counterparts in paid roles. Finally, an unpaid internship may not signal the same level of prestige to future employers and would thus not give a candidate the same competitive edge in the labor market.

Uncovering the true underlying mechanism (or the relative strengths of each of the three) has important policy implications. If the benefit of internships comes from enhanced skill acquisition, as would be predicted by human capital theory (Becker, 1964; Mincer, 1974) or the literature on high-impact practices (Kuh et al., 2008), broadening participation would lead to higher post-completion earnings for all students who had paid internships, commensurate with their higher productivity and value to employers. In this case, policymakers should work to ensure that as many students as possible are able to participate in paid internships, with a special focus on those who have the lowest participation rates. There is some evidence for this mechanism, but it is the weakest of the three we examine.

If the key mechanism is the development of professional connections, the implications are less straightforward. On one hand, if overall salaries for entry-level workers do not change, more paid internships and thus strong professional connections for more students would not lead to elevated wages on average, since in this scenario there are a limited number of higher wage entry-level positions. At the same time, broadening internship participation might make access to higher wage entry-level jobs more equitable by leveling the playing field in terms of which students have access to strong professional networks (Di Meglio et al., 2022; Nunley et al., 2016). Furthermore, when employers are connected with a broader applicant pool, they should be able to increase the quality of their hires and thus increase productivity and wages. In this case, policymakers might seek to increase participation in paid internships both for equity reasons as well as enhancing the talent pool

Table 12 Co-ops and early career outcomes

Variable	1st year annual earnings	4th year annual earnings	Job requires degree	Career-related job
Co-op	2437.94	2030.11	0.17	−0.13
	(1500.28)	(2428.37)	(0.19)	(0.19)
Black	−968.78	−5184.00**	−0.08	−0.26
	(1273.88)	(1802.00)	(0.17)	(0.15)
Hispanic	−776.07	103.74	0.03	−0.05
	(1024.51)	(2211.12)	(0.17)	(0.15)
Female	−2313.24**	−6602.61**	−0.28*	−0.05
	(808.63)	(1446.59)	(0.11)	(0.10)
Parental education	−191.38	−195.91	0.05	0.01
	(195.48)	(401.15)	(0.03)	(0.03)
Income quartile	1351.81**	1595.54*	0.06	0.07
	(343.08)	(633.63)	(0.05)	(0.04)
Field of study (ref: Computer Science)				
Engineering & engineering technology	−2936.47	−452.56	0.4	−0.74*
	(3286.78)	(4247.05)	(0.30)	(0.29)
Biological & physical sciences	−17,986.77**	−19,231.31**	−0.58*	−0.72**
	(2849.07)	(3795.58)	(0.28)	(0.26)
General studies & other	−16,540.61**	−16,020.83**	−0.62	−0.78*
	(3818.52)	(5803.64)	(0.46)	(0.39)
Social sciences	−17,471.80**	−21,037.55**	−0.20	−0.98**
	(2801.03)	(3684.11)	(0.27)	(0.26)
Humanities	−20,504.60**	−23,738.73**	−0.58*	−0.93**
	(2887.46)	(3944.06)	(0.29)	(0.27)
Health care fields	−4740.40	−4103.31	−0.15	−0.02
	(2985.23)	(3961.69)	(0.29)	(0.28)
Business	−10,436.88**	−9356.87*	−0.35	−0.91**
	(2871.55)	(3908.84)	(0.27)	(0.26)
Education	−17,406.75**	−21,647.86**	0.43	−0.15
	(3051.85)	(3880.11)	(0.33)	(0.36)
Other applied	−17,621.26**	−16,012.28**	−0.31	−0.74**
	(2791.36)	(3917.87)	(0.27)	(0.26)
Bachelor's GPA	767.88	65.89	0.37**	0.13
	(910.71)	(1648.48)	(0.12)	(0.12)
High school GPA	−977.11*	−1347.64	0	0.04
	(441.72)	(724.22)	(0.06)	(0.05)
SAT scores	−0.26	1.57	−0.00	0
	(2.47)	(4.12)	(0.00)	(0.00)
Institutional selectivity	−442.52	−2628.63**	–0.25**	0.04
	(522.31)	(991.51)	(0.07)	(0.06)
Constant	48,557.67**	81,552.39**	−1.02	−0.03
	(4900.86)	(8295.71)	(0.63)	(0.57)
Observations (unweighted)	3890	3140	3890	3890

Table 12 (continued)

Variable	1st year annual earnings	4th year annual earnings	Job requires degree	Career-related job
Prob>F	0	0	0	0
R-squared	0.14	0.12	–	–

* $P \leq 0.05$, ** $P \leq 0.001$

Source: U.S. Department of Education, National Center for Education Statistics, 2016/20 Baccalaureate and Beyond Longitudinal Study (B&B:2016/20).

and helping their businesses to be more productive and competitive. This hypothesis has a somewhat stronger evidence base from our analysis.

The mechanism that seems to have the strongest evidence base in our analysis is that of symbolic capital. If the returns to paid internships are mainly because employers view them as a differentiator that signals higher abilities, broadening participation would dilute the signal and thus the value of a paid internship. Paid internships could become simply one more requirement or hurdle for would-be jobseekers, and, in the absence of ways to broaden participation, simply a way of perpetuating disadvantage.

For institutions and policymakers that wish to promote economic mobility and opportunity for all students, it will be critically important to ensure paid internships are accessible and attainable for all students even as the landscape for securing a paid internship becomes ever more competitive in the arms race among students to keep up and stand out in the job market. One challenge will be how to promote access for disadvantaged students in ways that do not run afoul of state and federal law. Examples could include coaching students on how to find and apply for internships, offering stipends to offset transportation and other costs associated with participating in internships, and offering schedule flexibility and credit for internships so students are able to balance internship participation with the demands of their coursework.

Another beneficial tactic that institutions can take is to help students better understand how to demonstrate the value of the non-internship experiences that they have, such as their on-campus and off-campus jobs, their student activities, and the projects they complete in their coursework. Often students do not understand the connection between these experiences and relevant skills that employers might be looking for, so coaching from supervisors, advisors, and faculty on how to represent these experiences on a resume and talk about them in a job interview could be very helpful.

More research is needed to better understand employer perceptions of work-based learning and how they incorporate work-based learning into their decisions around screening, hiring and compensation. This could help shed light on the relative importance of skill development, networking, and signaling. There is also more work to be done to understand the extent and the circumstances under which each type of work-based learning conveys skills, professional connections, or prestige. Understanding what employers are looking for and the extent to which various models of work-based learning deliver can help institutions enhance the value of a broader range of work-based learning experiences and focus public resources where they are likely to have the greatest benefit for both students and employers.

In addition, future research could examine how the relationship between work-based learning and career outcomes might vary or be conditional on student demographic or academic characteristics. For example, this study controlled for major, but future research

Table 13 Work-study and early career outcomes

Variable	1st Year Annual Earnings	4th Year Annual Earnings	Job Requires Degree	Career-Related Job
Work-Study	−990.17	−2231.96	0.24	−0.02
	(1075.58)	(2090.79)	(0.16)	(0.14)
Black	−964.93	−5010.48**	−0.12	−0.27
	(1215.37)	(1843.92)	(0.17)	(0.15)
Hispanic	−1274.44	−25.89	−0.02	−0.04
	(1022.75)	(2224.59)	(0.16)	(0.14)
Female	−1978.52*	−7135.80**	−0.30**	−0.04
	(790.93)	(1432.12)	(0.11)	(0.10)
Parental education	−148.41	−219.66	0.06*	0.02
	(189.89)	(396.13)	(0.03)	(0.03)
Income quartile	1377.38**	1185.42	0.05	0.09*
	(338.16)	(635.03)	(0.05)	(0.04)
Field of study (ref: Computer Science)				
Engineering & engineering technology	−4903.39	−5040.22	0.28	−0.89**
	(3153.07)	(4368.55)	(0.29)	(0.29)
Biological & physical sciences	−18,179.05**	−21,239.29**	−0.52*	−0.74**
	(2686.93)	(3820.59)	(0.26)	(0.25)
General studies & other	−17,579.83**	−19,659.66**	−0.64	−0.85*
	(3690.72)	(6034.99)	(0.46)	(0.38)
Social sciences	−17,821.82**	−22,728.86**	−0.27	−1.10**
	(2648.13)	(3758.63)	(0.26)	(0.25)
Humanities	−21,492.89**	−26,770.87**	−0.62*	−1.05**
	(2722.92)	(3934.15)	(0.28)	(0.26)
Health care fields	−4713.15	−6172.78	−0.18	−0.10
	(2842.48)	(3984.56)	(0.28)	(0.27)
Business	−10,969.31**	−10,611.38**	−0.27	−0.92**
	(2722.11)	(3961.86)	(0.26)	(0.25)
Education	−17,162.62**	−24,783.24**	0.25	−0.12
	(2876.52)	(4080.31)	(0.34)	(0.36)
Other applied	−17,997.15**	−17,878.57**	−0.24	−0.83**
	(2645.97)	(3986.47)	(0.26)	(0.25)
Bachelor's GPA	586.2	667.86	0.34**	0.13
	(882.13)	(1634.62)	(0.12)	(0.11)
High school GPA	−860.30*	−1367.40	0	0.04
	(431.54)	(710.08)	(0.06)	(0.05)
SAT scores	−1.54	1.52	−0.00	0
	(2.33)	(4.10)	(0.00)	(0.00)
Institutional selectivity	−536.63	−2461.19**	−0.21**	0.07
	(511.16)	(986.67)	(0.07)	(0.06)
Constant	49,700.56**	83,588.81**	−0.96	−0.11
	(4697.49)	(8092.89)	(0.61)	(0.55)
Observations (unweighted)	4000	3250	4000	4000

Table 13 (continued)

Variable	1st Year Annual Earnings	4th Year Annual Earnings	Job Requires Degree	Career-Related Job
Prob > F	0	0	0	0
R-squared	0.14	0.11	–	–

* $P \leq 0.05$, ** $P \leq 0.001$

Source: U.S. Department of Education, National Center for Education Statistics, 2016/20 Baccalaureate and Beyond Longitudinal Study (B&B:2016/20).

Table 14 Expected relationships between variables to test hypotheses of internship value

	First job secured through internship	First job related to internship	Paid internship
Human capital	Strong positive relationship	Strong positive relationship	No relationship or relatively weak relationship
Social capital	Strong positive relationship	No relationship or relatively weak relationship	No relationship or relatively weak relationship
Symbolic capital	No relationship or relatively weak relationship	No relationship or relatively weak relationship	Strong positive relationship

could look at the interaction between major, work-based learning experience, and earnings or career perceptions to see whether the relationships are stronger or weaker for certain fields of study. Similarly, future research could examine whether the relationship between paid internships and earnings is stronger or weaker for groups who have lower earnings on average, such as women, Black students, or first-generation students. Such a study would highlight whether paid internships have potential to close equity gaps in post completion outcomes.

In summary, this study strengthens prior literature by providing evidence that distinguishes between types of work-based learning, showing that paid internships are uniquely associated with stronger earnings and career outcomes. Further, this study demonstrates that most of the benefits of paid internships are direct, not just due to job matching, potentially due to their function as a signaling or screening method for employers. These findings also highlight the need for policymakers and institutions to prioritize and expand access to paid internships as a targeted strategy for improving economic mobility and advancing equity in postsecondary education.

Table 15 Testing mechanisms of internship value

Variable	1st year annual earnings	4th year annual earnings	Job requires degree	Career-related job
Paid internship	4849.82**	7555.18**	0.59**	0.69**
	(566.55)	(1060.19)	(0.07)	(0.07)
Found first job through internship	3452.25**	6842.86**	0.47**	0.56**
	(880.70)	(2060.25)	(0.10)	(0.11)
First job related to internship	1273.96**	1221.95*	0.16**	0.25**
	(273.88)	(516.46)	(0.04)	(0.04)
Black	−417.07	−3401.06**	0.02	−0.31**
	(742.44)	(1089.76)	(0.09)	(0.09)
Hispanic	−230.91	−33.93	0.1	−0.10
	(707.86)	(1283.76)	(0.09)	(0.09)
Female	−2316.00**	−6194.73**	−0.08	0.02
	(491.51)	(942.75)	(0.06)	(0.06)
Parental education	−33.92	267.71	0.04*	0.03
	(121.31)	(230.07)	(0.02)	(0.02)
Income quartile	1199.75**	1482.71**	0.03	0.06*
	(208.65)	(384.48)	(0.03)	(0.03)
Field of study (ref: Computer Science)				
Engineering & engineering technology	−3173.38	−13,628.19**	0.34*	−0.27
	(2102.26)	(4502.83)	(0.16)	(0.18)
Biological & physical sciences	−23,062.70**	−38,946.21**	−0.42**	−0.71**
	(1950.31)	(4327.83)	(0.15)	(0.17)
General studies & other	−21,174.04**	−32,768.20**	−0.43	−0.45
	(2557.33)	(5281.38)	(0.28)	(0.27)
Social sciences	−22,544.22**	−34,418.25**	−0.37*	−0.70**
	(1906.72)	(4391.42)	(0.15)	(0.17)
Humanities	−26,547.21**	−43,077.47**	−0.63**	−0.78**
	(1925.47)	(4293.54)	(0.16)	(0.17)
Health care fields	−10,212.93**	−22,260.14**	−0.37*	0.01
	(1986.69)	(4324.09)	(0.16)	(0.18)
Business	−15,905.22**	−22,941.57**	−0.24	−0.56**
	(1933.00)	(4354.39)	(0.15)	(0.17)
Education	−18,827.17**	−36,345.27**	0.35*	0.21
	(1963.91)	(4226.77)	(0.17)	(0.19)
Other applied	−22,250.32**	−34,491.94**	−0.35*	−0.60**
	(1883.89)	(4227.24)	(0.15)	(0.16)
Bachelor's GPA	1202.72*	909.69	0.32**	0.21**
	(575.41)	(1119.49)	(0.07)	(0.07)
High school GPA	−541.40	−1015.73*	−0.02	0.02
	(283.06)	(445.49)	(0.03)	(0.03)
SAT scores	5.09**	14.35**	0	0.00*
	(1.55)	(3.11)	(0.00)	(0.00)
Institutional selectivity	−1066.39**	−3834.76**	−0.25**	−0.05
	(362.65)	(674.22)	(0.04)	(0.04)
Constant	42,293.12**	76,021.64**	−1.49**	−1.01**
	(3126.04)	(6564.13)	(0.35)	(0.35)

Table 15 (continued)

Variable	1st year annual earnings	4th year annual earnings	Job requires degree	Career-related job
Observations (unweighted)	11,960	10,040	11,960	11,960
Prob > F	0	0	0	0
R-squared	0.19	0.17	–	–

* $P \leq 0.05$, ** $P \leq 0.001$

Source: U.S. Department of Education, National Center for Education Statistics, 2016/20 Baccalaureate and Beyond Longitudinal Study (B&B:2016/20).

Table 16 Mediation analysis

First-year annual earnings	Coefficient	Std. Err	z	P > z	[95% Conf. Interval]
Indirect association Paid internship	**1182.54	264.61	4.47	0.000	[663.92, 1701.16]
Direct association Paid internship	**4142.42	715.24	5.79	0.000	[2740.57, 5544.27]
Total association Paid internship	**5324.96	681.89	7.81	0.000	[3988.49, 6661.43]

** $P \leq 0.001$

Number of observations (unweighted) = 5070.

Outcome model: Linear Mediator model: Logit.

Mediator variable: First job found through internship Treatment type: Binary

Source: U.S. Department of Education, National Center for Education Statistics, 2016/20 Baccalaureate and Beyond Longitudinal Study (B&B:2016/20).

Table 17 Mediation analysis

First job requires degree	Coefficient	Robust Std. Err	z	P > z	[95% Conf. Interval]
Indirect association Paid internship	**0.019	0.006	3.32	0.001	[0.008, 0.030]
Direct association Paid internship	**0.123	0.017	7.38	0.000	[0.090, 0.155]
Total association Paid internship	**0.142	0.016	8.88	0.000	[0.111, 0.173]

** $P \leq 0.001$

Number of observations (unweighted) = 5070 Outcome model: Logit.

Mediator model: Logit.

Mediator variable: First job found through internship Treatment type: Binary

Source: U.S. Department of Education, National Center for Education Statistics, 2016/20 Baccalaureate and Beyond Longitudinal Study (B&B:2016/20).

Appendix: Tables

Tables 18, 19, 20, 21 and 22.

Table 18 Percentage distribution of 2015–16 bachelor's degree earners, by demographic, enrollment, and employment characteristics in 2020

Characteristic	Percentage (%)
Biological sex (at birth)	
Female	57.4
Male	42.6
Race/ethnicity	
White, not Hispanic or Latino	65.5
Hispanic or Latino, of any race	12.5
Black, not Hispanic or Latino	10.1
Asian, not Hispanic or Latino	7.8
Two or more races, not Hispanic or Latino	3.3
American Indian or Alaska Native, not Hispanic or Latino	0.5
Native Hawaiian or Other Pacific Islander, not Hispanic or Latino	0.3
Highest education attained by either parent as of 2015–16 academic year	
Graduate or professional degree	29.6
Bachelor's degree	28.5
Some postsecondary education or subbaccalaureate credential	25.4
High school diploma or less	16.5
Major for 2015–16 bachelor's degree	
Business	19.8
Other applied	15.1
Social sciences	14.7
Health care fields	11.5
Humanities	10.1
Engineering and engineering technology	6.2
Education	4.6
Computer and information sciences	3.5
Biological and physical sciences	3.5
General studies and other	2.3

Source: U.S. Department of Education, National Center for Education Statistics, 2016/20 Baccalaureate and Beyond Longitudinal Study (B&B:2016/20).

Table 19 Participation rates in work-based learning experiences by parents' highest education level

Parents' highest education	Paid internship (%)	Unpaid internship (%)	Co-op (%)	Practicum (%)	Work study (%)
Do not know	4.9	36.3	6.0	19.1	4.2
Did not complete high school	17.9	26.8	10.4	15.1	6.6
High school diploma/GED	20.0	28.5	7.9	15.0	8.5
Vocational/technical training	24.3	25.1	13.5	16.8	10.0
Associate's degree	27.0	31.0	9.0	15.5	8.9
Some college	24.1	28.2	11.0	14.7	10.3
Bachelor's degree	31.9	32.9	10.5	15.1	10.1
Master's degree	33.0	30.7	10.5	14.7	10.3
Doctoral degree	36.0	34.3	11.9	17.2	9.0
Professional degree	36.5	35.5	7.8	15.0	10.3
Total	28.7	30.8	10.2	15.2	9.7

Source: U.S. Department of Education, National Center for Education Statistics, 2016/20 Baccalaureate and Beyond Longitudinal Study (B&B:2016/20).

Table 20 Participation rates in work-based learning experiences by income quartile

Income quartile	Paid internship (%)	Unpaid internship (%)	Co-op (%)	Practicum (%)	Work study (%)
Bottom quartile	26.0	33.6	10.8	15.4	12.7
Lower-middle quartile	26.7	33.1	12.3	16.4	13.0
Upper-middle quartile	28.6	30.1	10.0	15.1	10.2
Top quartile	31.6	28.6	8.9	14.3	5.7
Total	28.6	31.0	10.3	15.2	9.9

Source: U.S. Department of Education, National Center for Education Statistics, 2016/20 Baccalaureate and Beyond Longitudinal Study (B&B:2016/20).

Table 21 Participation rates in work-based learning experiences by institutional selectivity

Institutional selectivity	Paid internship (%)	Unpaid internship	Co-op experience	Practicum	Work study
Very selective	39.1	37.3	9.9	11.6	13.6
Moderately selective	27.3	30.3	10.8	16.5	9.7
Minimally selective	17.3	23.4	11.0	19.7	6.3
Open admission	12.0	20.9	7.3	12.7	1.5
Total	28.6	31.0	10.3	15.2	9.9

Source: U.S. Department of Education, National Center for Education Statistics, 2016/20 Baccalaureate and Beyond Longitudinal Study (B&B:2016/20).

Table 22 Total participation in work-based learning experiences

No work-based learning	Paid internship only	Unpaid internship only	Co-op experience only	Practicum only	Work study only	More than one type of work-based learning
7870	2000	2550	470	1300	490	5650

Source: U.S. Department of Education, National Center for Education Statistics, 2016/20 Baccalaureate and Beyond Longitudinal Study (B&B:2016/20).

Declarations

Competing interests The authors declare no conflicts of interest related to this work to disclose.

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